

Machine Learning and IoT Based Automated Remote Health Monitoring and Alert System

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Abstract—The rapid evolution of Internet of Things (IoT) technologies has significantly transformed healthcare monitoring, enabling continuous, real-time, and remote patient observation. This paper presents the design and implementation of a low-cost, portable IoT-based patient health monitoring system. The system integrates multiple biomedical sensors to measure vital parameters such as ECG, SpO₂, heart rate, body temperature, and fall detection into a portable, cost-effective, and scalable platform. The system employs ESP32 as the central controller and supports both local (OLED/TFT displays) and remote (cloud-based dashboards and Telegram alerts) visualization. Safety features such as toxic gas detection and smoke detection as well as ambient temperature and humidity are incorporated to enhance patient well-being, particularly for elderly and chronically ill individuals. Furthermore, a machine learning model deployed via Google Colab performs automated ECG classification, achieving a validation accuracy of 94.23%, enabling early detection of cardiac anomalies. Experimental results demonstrate high accuracy across physiological parameter measurements (>97%) and fast alert delivery (<5 seconds), confirming the system's potential as a reliable solution for home-based and rural healthcare applications.

Keywords—IoT, Machine Learning, Cloud Connectivity, Constant Health Monitoring, ECG, Automated Dual Notification

I. INTRODUCTION

This The concern for rapid and reliable health checkups for elderly people and people with chronic illness is one of the main challenges of medical science. According to the World Health Organization (WHO), chronic diseases are responsible for approximately 75% of all global deaths [1], and cardiovascular diseases alone account for 19.8 million deaths annually worldwide[2]. As the number of people having such conditions is increasing, the demand for accurate health monitoring system is on the rise. On the other hand, a huge population, nearly 59% of the people of Bangladesh, live in the rural areas[3], who do not have access to proper healthcare systems. Thus, they do not get the opportunity to have immediate diagnosis and detection of diseases during an emergency. Internet of Things (IoT) has opened a new dimension in this regard. It opened a door for performing remote checkups with accuracy that was thought to be not

possible at one time. Thanks to the advancement in sensors, which are more reliable and wearable and the advancement in the field of cloud technology we can now build simple and easy remote health monitoring systems, that are faster, accurate and cost-effective.

On the other hand, the groundbreaking advancement in the field of data science, particularly Machine Learning (ML) has given an opportunity to analyze the data collected from the sensors and evaluate the patient's health and perform quick response if required. Thus, a cloud-based framework backed up by ML based evaluation can provide a sophisticated yet simple approach to monitor a patient's health and perform immediate response in case any irregularity is detected in the collected sensor data.

In this study, we designed an IoT based remote health monitoring system with the integration of ML based health condition evaluation and automated notification system. The system constantly measures all vital signs of the patient and securely sends them to the cloud to be evaluated. The system can also record ECG signals and fall detection of the patient. These two records of the patients ensure the early detection of diseases like cardiac arrests, triggering of epilepsy, and many other cardiovascular diseases along with the ability to constantly monitor the health condition of elderly people and people with chronic illness. An ML based ECG evaluation system is designed to ensure accurate reading of the ECG signals and take steps accordingly. The system also comes with a built-in gas detector to detect harmful gases for people having asthma, emphysema and other respiratory diseases. If an event of any irregular condition detection within the patient either from the sensors or from the ML based ECG report, the system will send a notification to the user and the medical officers. It will also send the current report along with the patient's status and the environment he is currently in, so that quick and immediate response can be provided by the medical professionals. Thus, allowing them to monitor the patient from distance and provide treatment remotely and instantly improving the chance of survival of the patient in case of an emergency making healthcare accessible to all efficiently.

II. LITERATURE REVIEW

Healthcare monitoring has undergone a major transformation with the adoption of IoT, offering connected, automated, and scalable solutions for patients and caregivers. Hashimah et al. [4] proposed an IoT-based healthcare monitoring system using pulse, infrared thermometer, and DHT11 sensors connected to Arduino Mega. Data were uploaded to ThingSpeak, enabling remote access for physicians. Kumar et al. [5] emphasized real-time health tracking using ECG, heart rate, and temperature sensors integrated with NodeMCU and cloud servers. Mahajan and Birajdar [6] focused on chronic disease management, using Particle Photon as an IoT gateway for measuring temperature, SpO₂, and heart rate. MQTT protocols supported efficient data transfer, and visualization was facilitated through the Adafruit server. The system was designed for scalability, affordability, and remote accessibility, making it suitable for long-term home care and early detection of conditions like cardiac arrest and asthma. Another study [7] develops innovative wearable solutions such as smart gloves have been introduced for elderly care, combining fall detection (MPU6050), gesture recognition, and SpO₂ monitoring, with alerts sent via IFTTT to caregivers. This system bridge gaps in eldercare by addressing fall-related mortality and improving independence.

III. PROBLEM FORMULATION

Though there have been significant advancements in IoT-based health monitoring system, most prior work remains limited in scope as they focus on only a subset of patient health and safety needs. Some works primarily monitor basic physiological parameters such as heart rate, temperature, or SpO₂, whereas others do work only on only specialized like ECG monitoring. Some health monitoring systems only detect fall detection for elderly people. So, they do not provide a comprehensive view of the patient's overall health status. Besides, many systems depend solely on remote dashboards causing delay in timely intervention during emergencies. Therefore, the core problem is to design an integrated, reliable, and secure IoT-based health monitoring system that:

- Captures multi-parameter biomedical data with high accuracy.
- Provides real-time visualization locally and remotely.
- Issues timely alerts for abnormal readings to enable rapid intervention.
- Remains portable, cost-effective, and scalable for deployment in home-care and rural environments.

IV. PROPOSED SYSTEM

Our proposed system addresses the gaps stated in problem formulation section by integrating multiple biomedical sensors (e.g., ECG, SpO₂, temperature, and motion) into a single portable platform. It enables simultaneous monitoring of critical health parameters. Our system incorporates real-time data visualization both locally (by LCD/OLED display) and remotely (via a cloud platform) for the enhancement of usability and accessibility. Additionally, the design emphasizes portability and cost-effectiveness keeping in mind the suitability for home-based and rural healthcare settings where access to advanced facilities is limited. The system also aims to minimize false positives, ensure data accuracy. Thus, our proposed system reduce the need for continuous physical

supervision by combining local processing, cloud integration, and automated alerts.

V. SYSTEM ARCHITECTURE

A. Block Diagram of Total System

First, The proposed system is built around an ESP-32 Main Control Unit that integrates health monitoring, environmental sensing, and real-time alerting. As shown in the (Fig. 1), our health monitoring system consists of two main modules- a Main unit and an Extension Unit. The Main Unit handles Health Monitoring (heart rate, SpO₂, and body temperature) and Ambient Monitoring (ambient temperature, humidity, fire detection, and toxic gas detection) to ensure both patient health and environmental safety. The Extension Unit adds ECG Monitoring for detailed cardiac analysis and Fall Detection to improve elderly patient safety.

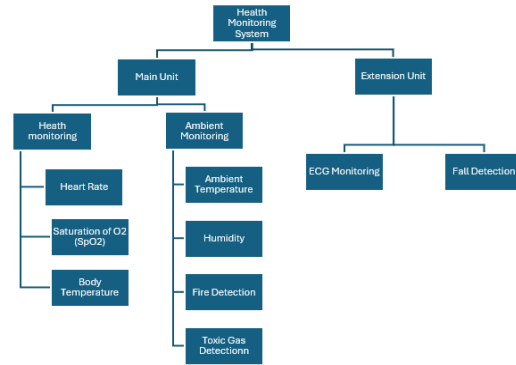


Fig. 1. Block Diagram of Health Monitoring System

B. SpO₂ and Heart Rate Monitoring System

The system utilizes the MAX30102 for the measurement of a patient's blood oxygen saturation (SpO₂) and heart rate (BPM) in real time. The MAX30102 optical sensor measures SpO₂ and heart rate using photoplethysmography (PPG). Red and infrared LEDs emit light into the skin, and reflected signals are processed to compute pulse rate and oxygen concentration. These health parameters are continuously monitored and displayed on the onboard OLED display. Besides, the results are sent to Blynk App and alerts are sent via Telegram bot when SpO₂ is low or heart rate is high, allowing remote caregivers or family members to stay informed via their mobile devices.

C. Body Temperature Monitoring System

Using the DS18B20 sensor, the system tracks the patient's body temperature with high precision in Fahrenheit. Results are instantly shown on the OLED display and shared via Telegram, ensuring continuous monitoring both nearby and remotely. Continuous body temperature tracking supports early detection of fever, infection, or hypothermia, enabling prompt medical intervention.

D. Temperature and Humidity Monitoring System

Maintaining optimal environmental conditions is crucial for patients with respiratory illnesses or chronic health conditions. The system uses DHT11 sensor for this purpose. DHT11 sensor records ambient temperature and relative humidity. Readings are displayed locally and transmitted to remote platforms for continuous monitoring.

E. Safety Monitoring System

The system integrates MQ2 gas sensor to detect the presence of smoke, LPG, and other toxic gases commonly found in domestic environments. When the sensor identifies abnormal gas levels beyond a safe threshold, the system immediately triggers an alert. Here the system also uses dual-alert system which ensure that both local and remote users are instantly informed. enables. By adding this safety layer, swift action can be taken in case of potential fire hazards or gas leaks. Thus, the system not only monitors patient health but also actively safeguards the surrounding environment.

F. Fall Detection System

The MPU6050 accelerometer and gyroscope module is used to monitor patient movement in three dimensions. The system calculates total G-force and detects abnormal deviations by continuously measuring linear acceleration and angular velocity. Falls in any direction—forward, backward, left, or right—are identified using a calibrated threshold. The system activates local alerts through LEDs and buzzers while simultaneously sending notifications via a Telegram bot to caregivers on fall detection. This ensures immediate response in emergencies, making it particularly valuable for elderly and high-risk patients.

G. ECG Monitoring System

The system includes advanced ECG monitoring using the AD8232 module, which captures and displays the patient's ECG signal in real time on the onboard TFT display. Besides, for deeper analysis, an ML model is deployed which is discussed in later section. Thus the system provides automated comments on ECG quality, indicating whether the signal reflects a healthy cardiac condition or potential anomalies. It uses a decision making algorithm for deciding about the cardiac condition and sending a notification to the caregiver. The flowchart (Fig. 2) shows the exact paths for condition detection and notification logic.

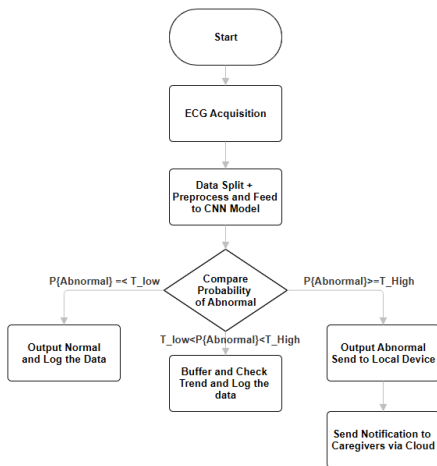


Fig. 2. Flowchart for Alerting paths and Caregiver Notification Logic

VI. IMPLEMENTATION AND RESULTS

A. Hardware Implementation

Basic Health Monitoring Unit: One ESP32 module was mounted on a veroboard which was connected to four sensors for reading basic vital signs and monitoring of health status

(Fig. 3). These sensors were MAX30102 Pulse Oximeter for monitoring pulse rate and blood oxygen level, a DS18B20 temperature sensor for measuring body temperature, a DHT11 module for measuring ambient temperature and humidity used as reference for the body temperature reading, MQ2 gas detector for measuring any toxic gas in the surroundings. The readings of the sensors are shown to an OLED display (Fig. 14) which is connected to the ESP32 module. Also if the readings are unusual then a notification is sent to the phone (Fig. 15). The sensors produce sufficient accurate results and combining them together can ensure the proper detection and monitoring of all vital signs of the user.

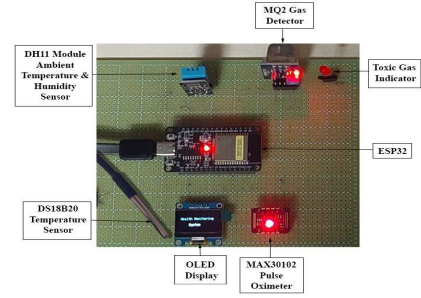


Fig. 3. Basic Health Monitoring Unit

ECG and Fall Detection Unit: This is an extension unit of the system which consists of ECG sensors and fall detectors for elderly people and people with chronic heart disease (Fig. 4). An ESP32 is connected to a AD8232 ECG sensor which is connected to three probes for detecting the ECG signal and the ESP32 module is also connected to a MPU6050 fall detector which is a gyroscope sensor. When ECG signal is read the signal is shown in a TFT display in real time (Fig. 4) and it is also being sent to the cloud for analyzing whether it is a normal ECG or not using machine learning approach (Fig. 5). The fall detector measures the orientation of the user and if he starts falling then it sends a notification to the phone (Fig. 15) and the fall detector LED turns on as well.

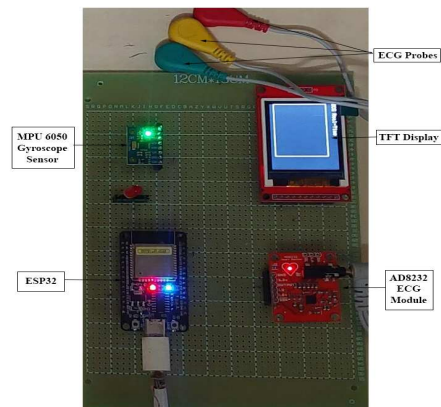


Fig. 4. ECG and Fall Detection Unit

B. ML Model Implementation and Cloud Based ECG Detection

The readings of the ECG sensor are sent to a Google Sheet which is linked to a Google Colab script. The colab script consists of an ML model trained on detecting abnormal and

normal ECG reports from the “ptbdb_abnormal.csv” and “ptbdb_normal.csv” datasets from Kaggle [8]. The model reads the values from Google Sheet and then performs inference based on its training and then gives the output whether it is normal or abnormal and it also plots the ECG signal (Fig. 5).

ML Model Architecture: The ML model was deployed in the following manner:

- Three Conv1D layers with 16, 32 and 64 filters, kernel size 5 and each followed by ReLU activation.
- Two Maxpooling1D layers of pool size 2 after the first two convolutional layers.
- One Global Average Pooling layer after the final convolutional layer.
- A single fully connected neuron with sigmoid activation to produce probability $p(\text{Abnormal})$.
- The model was compiled using ‘adam’ optimizer and ‘binary cross-entropy’ was used as loss function.

Dataset Splits and Evaluation Protocol: The dataset from Kaggle [8] contains 14,552 entries with 187 features per entry. The dataset was partitioned in the following manner:

- All beats are first combined and stratified into
 - 80% training pool (11, 641 entries)
 - 20% held-out test set (2,911 entries)
- On the training pool, a 5-fold stratified cross-validation was performed to report mean $\pm$ standard deviation.
- For the final model, the training pool was re-split into:
 - 90% training
 - 10% internal validation

The model was trained with early stopping and evaluated once on the fixed held-out test set.

Patient-Wise Separation: Since the publicly available PTBDB heartbeat dataset from Kaggle [8] does not include subject identifiers, we were unable to enforce a strict patient wise data split. Consequently, our train-test division is performed on a heartbeat-wise basis based on the entries, stratified by class labels to maintain balance. We acknowledge this as a potential limitation due to possible patient level data leakage and recommend future studies to perform subject level splitting using the original PTBDB records.

Pre-Processing: Using only the train pool statistics, each feature was standardized as follows:

$$x' = \frac{x - \mu_{train}}{\sigma_{train} + 10^{-8}}$$

The same μ_{train} and σ_{train} were applied to validation and test sets. Inputs were reshaped to (187, 1) for Conv1D layers.

Data Augmentation (training only):

- Mild amplitude scaling was done with a random factor in [0.9, 1.1]
- Additive Gaussian Noise with $\sigma = 0.02$ was used.

These were only applied to training batches.

The model had an accuracy of 94.23% (Fig. 6).

The classification report (Fig. 6) and the confusion matrix (Fig. 7) confirm the accuracy of the model. The training and validation accuracy and loss plots (Fig. 13) show that the model has learnt important features throughout its training. A 5-fold stratified cross-validation on a held-out test set was done which yielded an accuracy of 0.9424 ± 0.0086 , ROC-AUC of 0.9836 ± 0.0025 and PR-AUC of 0.9922 ± 0.0017 as mean $\pm$ SD (Fig. 8).

The calibration curve (Fig. 9) shows that probabilities are reasonably aligned with actual event frequencies with some overconfidence in the mid-range bins. The Brier score of 0.0397 (Fig. 9) supports good but not perfect calibration. Based on the evaluation on the held-out test set, the Confusion Matrix (Fig. 10) shows high correct classification with a ROC Curve (Fig. 11) indicating excellent discriminative capability and a Precision-Recall Curve (Fig. 12) confirming highly reliable detection. All these demonstrate that the model has a strong overall performance.

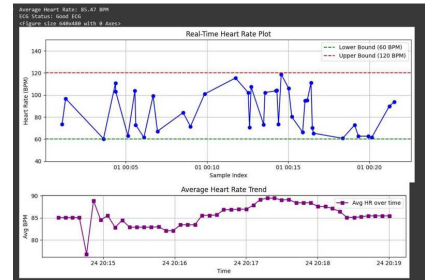


Fig. 5. Cloud Based ECG Report Using ML Approach

Classification Report:				
	precision	recall	f1-score	support
0.0	0.9234	0.8648	0.8927	809
1.0	0.9489	0.9724	0.9605	2102
accuracy				
macro avg	0.9362	0.9182	0.9266	2911
weighted avg	0.9418	0.9423	0.9417	2911
Validation Accuracy: 0.9423878735829612				

Fig. 6. Classification Report

Confusion Matrix		
True Label	Normal	Abnormal
	699	110
Predicted Label	Normal	Abnormal
	58	2044

Fig. 7. Confusion Matrix

CV Summary (mean $\pm$ SD)	
accuracy:	0.9424 ± 0.0086
roc_auc:	0.9836 ± 0.0025
pr_auc:	0.9922 ± 0.0017

Fig. 8. A 5-Fold Stratified Cross-Validation Summary on a Held-Out Test Set

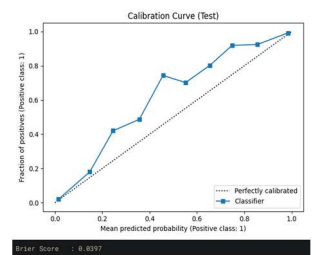


Fig. 9. Calibration Curve

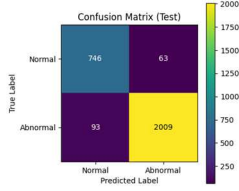


Fig. 10. Confusion Matrix on held-out test set

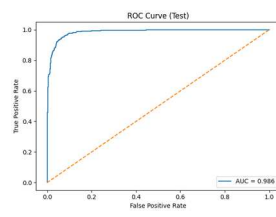


Fig. 11. ROC Curve

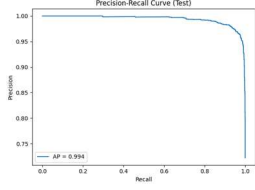


Fig. 12. Precision-Recall Curve

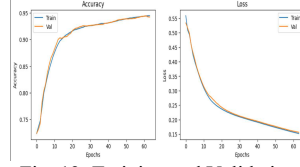


Fig. 13. Training and Validation Accuracy and Loss Plots over epoch

Model Card:

Model: 1D CNN for binary ECG classification between normal and abnormal using 187 samples with single sigmoid output.

Intended Use: Decision maker for early detection of abnormal cardiac activity to send notification alerts to caregivers. It is not a stand-alone diagnostic tool or replacement for expert interpretation.

Data Sources: PTBDB-derived ECG Heartbeat Categorization Dataset [8] from Kaggle.

Limitations:

- Patient-level leakage could not be excluded due to not having patient ID in the dataset.
- Does not detect the exact type of cardiac disease. Outputs only binary label.
- Sensitive to noisy data.

Failure Modes:

- Occasional false alarms
- Risk of false detection for data outside the training data domain

C. System and Model Deployment and Result Collection

The system combined with various sensors can monitor vital health signs almost accurately and can cross check by ensuring that multiple vital signs are failing simultaneously. The values are always displayed on the OLED display (Fig. 14). The ECG is shown in the TFT display and the Colab script sends output whether it is a normal or abnormal ECG (Fig. 5). If any vital signs fail, the system will send a notification to the user and the mentioned contacts via cloud services (Fig. 15). All the data can always be read from the dashboard of an app (Fig. 16). Thus, the model can accurately monitor vital information and detect health issues and can trigger notifications to notify the user of any unusual vital signs.



Fig. 14. Vital Information Displayed on the OLED display

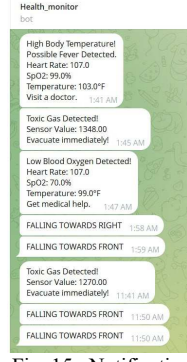


Fig. 15. Notification for Abnormal Readings



Fig. 16. App Dashboard

VII. RESULT ANALYSIS

This part shows the experimental data derived from the IoT-based smart health monitoring system, followed by an evaluation of its accuracy, dependability, and performance in relation to conventional medical instruments.

TABLE 1: Temperature Reading

Sample	Temperature (Infrared thermometer) (F)	Normal body temperature	Temperature (proposed system) (F)	Accuracy
01	98.57	98.6	98.24	99.66%
02	99.42	98.6	99.06	99.63%
03	98.2	98.6	99.01	99.18%
04	101.27	98.6	101.78	99.5%
05	96.02	98.6	96.47	99.54%

TABLE 2: SpO2 Level Reading

Sample	Pulse Oximeter reading	Normal oxygen saturation level	Reading (Proposed IOT-based system)	Accuracy
01	94.77	96-99	95.04	99.7%
02	92.48	96-99	94.25	98.09%
03	97.45	96-99	94.76	97.24%
04	95.24	96-99	97.78	97.46%
05	97.34	96-99	96.85	99.5%

From Table-1 and Table-2 we can have an accuracy of above 97%. From Fig. 6 and Fig. 8, the CNN model achieved an overall accuracy of 94.23% confirming high reliability in ECG data analysis also. The system also offers real-time alerting mechanisms both through the app dashboard and Telegram notifications. This ensures immediate awareness. This system ensures alert delivery within 5 seconds.

Baseline Comparison:

The originally deployed ECG model uses the same CNN architecture with a single 80/20 split. It had an accuracy of 94.23% without cross-validation, probability calibration and detailed error analysis. The revised model with the cross validation on held-out test data yields an accuracy of $94.24\% \pm 0.86\%$ via a 5-fold stratified cross validation. The ROC-AUC is 0.9856 and PR-AUC is 0.9939 for the held-out test data. The number of true positive (TP) and false positive (FP) predictions are 2009 and 63 respectively over total 2072

abnormal alerts from the confusion matrix (Fig. 10). This yields-

$$\text{False Positive Rate} = \frac{FP}{TP + FP} = \frac{63}{2072} = 3.04\%$$

The number of false negative (FN) predictions is 93 (Fig. 10) which yields-

$$\text{Missed Abnormal Rate} = \frac{FN}{TP + FN} = \frac{93}{2102} = 4.42\%$$

These two metrics confirm that the system has a very low error rate in detecting abnormal conditions while missing only a few events of having abnormality. Thus, the model is highly accurate in most cases for providing proper decisions.

Deployment Constraints:

- ESP-32 has limited memory and so the CNN is deployed in a lightweight form which was executed on a cloud.
- If network is not available, the ECG will not be evaluated but the data will be queued to upload when network becomes available next time.

Security and Privacy:

- All remote communications are configured to HTTPS.
- Device identity and API access are protected via unique keys. Only authenticated endpoints can read patient data.
- Necessary minimal data are transmitted. Rest of the data are processed on the local hardware.

TABLE 3: Cost analysis

Serial	Type	Component name	Cost(BDT)	Cost(USD)
01	Base Unit	ESP32 NodeMCU Dev Board	380	3.12
02		MAX30102 Pulse & SpO2 Sensor	150	1.23
03		DS18B20 Temp Sensor	120	0.99
04		DHT11 Temp & Humidity	100	0.82
05		MQ2 Gas/Smoke Sensor	100	0.82
06		OLED SSD1306 (0.96")	210	1.73
07		Veroboard	200	1.64
		Total Base Unit Cost	1260	10.35
08	Extended Unit	ESP32 NodeMCU Dev Board	380	3.12
10		AD8232 ECG Sensor	600	4.93
09		MPU6050 Accelerometer	180	1.48
11		TFT ST7735 Display	680	5.57
		Total Extended Unit Cost	1840	15.10
		Total Cost	3100/- BDT	25.45 USD

Per unit cost of the total prototype with the extension unit = 3100 BDT(25.45 USD). Per unit cost of the prototype without the extension unit (e.g. ECG and Fall Detector and related components) = 1260 BDT (10.35 USD) .This is much lower than most fitness bands and watches available in the market with these features.

CONCLUSION

In conclusion, this study proposes an IoT based portable platform for smart health monitoring system which monitors vital signs such as heart rate, SpO₂, body temperature, and ECG, while also incorporating fall detection and toxic gas sensing for enhanced patient safety. In case of abnormalities, real-time alerts are sent through mobile applications and notifications to ensure timely intervention. A 1D-CNN model based on machine learning was used for cloud-based ECG classification. It shows a validation accuracy of 94.23%. The suggested system provides a more complete solution than earlier efforts by integrating ML based diagnostic support, real-time alerting, and multi-sensor monitoring. Future improvements will focus on expanding the range of physiological parameters and incorporating advanced AI for predictive health analytics.

REFERENCES

- [1] World Health Organization, "Noncommunicable diseases," WHO, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases>
- [2] World Health Organization, "Cardiovascular diseases (CVDs)," WHO, 2023. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-cvds>
- [3] The World Bank, "Rural population (% of total population) – Bangladesh," World Bank Data, 2023. [Online]. Available: https://data.worldbank.org/indicator/SP.RUR.TOTL.ZS?locations=B_D
- [4] H. Hashim, S. F. B. Salihudin and P. S. M. Saad, "Development of IoT Based Healthcare Monitoring System," 2022 IEEE International Conference in Power Engineering Application (ICPEA), Shah Alam, Malaysia, 2022, pp. 1-5, doi: 10.1109/ICPEA53519.2022.9744712.
- [5] A. Kumar, U. Chauhan, M. Ansari, D. Sharma, T. Bahar and M. Kumar, "IoT Based Smart Health Monitoring System," 2024 1st International Conference on Advances in Computing, Communication and Networking (ICAC2N), Greater Noida, India, 2024, pp. 587-592, doi: 10.1109/ICAC2N63387.2024.10895618.
- [6] S. Mahajan and A. M. Birajdar, "IOT based Smart Health Monitoring System for Chronic Diseases," 2019 IEEE Pune Section International Conference (PuneCon), Pune, India, 2019, pp. 1-5, doi: 10.1109/PuneCon46936.2019.9105717.
- [7] M. F. Ahmed *et al.*, "Internet of Things Based Health Monitoring System Using Smart Glove," 2024 IEEE 3rd International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON), Dhaka, Bangladesh, 2024, pp. 282-286, doi: 10.1109/RAAICON64172.2024.10928501.
- [8] ECG Heartbeat Categorization Dataset. (2018, May 31). Kaggle. <https://www.kaggle.com/datasets/shayanfazeli/heartbeat>